KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2024
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

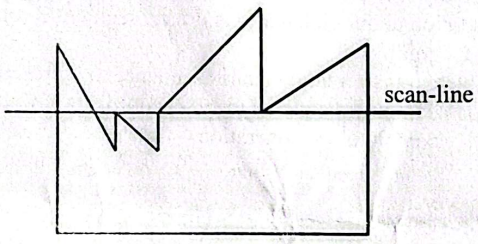
N.B. i) Answer ANY THREE questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer ANY THREE questions from this section in Script A)

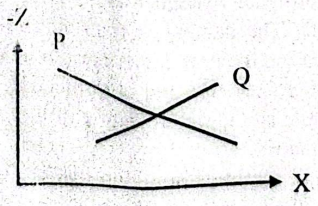
1. a) Suppose, you are designing a CRT monitor for displaying realistic scenes. Choose the appropriate combination of display type (vector/raster) and color production method (beam penetration/shadow mask) for the task and justify your choice. (10) [CO1]
   b) A line segment P_0P_1 is defined by the implicit equation $5x - 8y + 4 = 0$, where the end points are constrained such that $y_{start} = 8$ and $y_{end} = 13$. Using Bresenham's line drawing algorithm, determine the internal points. (12) [CO2]
   c) A graphics system renders two line segments: $L1$ $(-1, 7)$ to $(11, 1)$ and $L2$ $(5, 4)$ to $(9, 12)$. The visible area is restricted to a square shaped clipping window with $(x_{min}, y_{min}) = (1, 1)$ and $(x_{max}, y_{max}) = (9, 9)$. Using Liang-Barsky line clipping algorithm, determine the portion of the line segments that are visible within the window. (13) [CO3]

2. a) Explain how standard ray tracer be modified to generate soft shadow. (08) [CO3]
   b) Apply the Scan-Line Fill algorithm to demonstrate how the given polygon is filled along the specific scan-line. (10) [CO2]

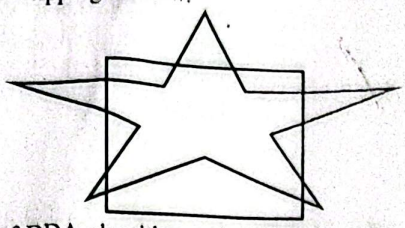


   c) A texture T of size 512×512 is mapped onto a planar surface. You are given a normalized coordinate point on that surface as $P(u, v) = (0.4, 0.6)$. Determine the corresponding texture coordinate $T(x, y)$ to be mapped on P and compute the final color at this point using GL_LINEAR texture filtering method. Given the color of the texels: $T(x, y) = (123, 205, 75)$, $T(x + 1, y) = (200, 150, 100)$, $T(x, y + 1) = (80, 220, 130)$, and $T(x + 1, y + 1) = (160, 90, 240)$. (17) [CO3]

3. a) Explain how the ray tracing method integrates the process of projection, scan conversion and shadow generation. (10) [CO2]
   b) A fractal is generated from an equilateral triangle. In each iteration, every side is replaced using the Koch snowflake construction, and each new edge is transformed into a Cantor edge of the same level. Now, draw the third-generation of the fractal and calculate the dimension of the structure. (12) [CO3]
   c) Given edge-on polygons P and Q in the $(x, -z)$ plane with overlapping (x, y) extents and assuming the view port looks towards the negative z axis. Apply depth sorting algorithm to examine the tests on the polygons and state the final rendering order of them. (13) [CO2]



4. a) Apply Weiler-Atherton polygon clipping algorithm to clip the star shaped polygon using the rectangle as a clipping window. (10) [CO1]



b) State the limitations of DDA algorithm and explain how Bresenham's line drawing algorithm overcomes them? (07) [CO1]

c) Consider a 3D scene where a point light source is located at the center of a sphere of radius 3 with center $C = (-1, 7, -4)$. A light ray emitted from the source intersects the sphere at point p, then strikes a triangle defined by $v_0 = (-5, 1, -5)$, $v_1 = (1, -1, -4)$, $v_2 = (0, 15, -7)$ with approximate surface normal $N = -1.04i + 1.77j + 9.79k$ at point q, and reflects towards the camera at origin $s = (0, 0, 0)$ passing through an image plane at depth 2. Trace the equivalent view ray from camera to the direction $d = (-1, 2, -2)$ and determine the intersection points p and q. (18) [CO3]

SECTION B

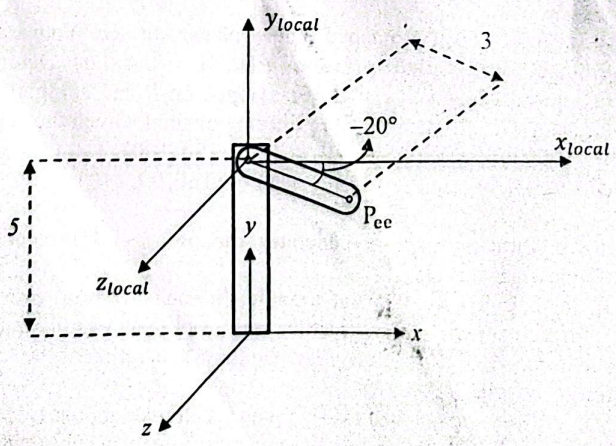
(Answer **ANY THREE** questions from this section in Script B)

5. a) Proof that 3D rotation matrix can be written as $R = \mathbf{I}\cos\theta + \sin\theta\,\mathbf{U} + (1 - \cos\theta)\mathbf{u}\mathbf{u}^T$, where $\mathbf{u}$ is unit axis of rotation, θ = angle of rotation, and $\mathbf{U}$ is skew symmetric matrix satisfying $\mathbf{Uv} = \mathbf{u} \times \mathbf{v}$. Be noted $\mathbf{U}^2 = \mathbf{u}\mathbf{u}^T - \mathbf{I}$. (18) [CO1]

b) Consider the following figure showing a 2 segment robotic arm fixed at origin. Segment 1 has length 5 and placed along y-axis. Segment 2 has length 3, attached to the end of segment 1, and initially it is oriented $-20°$ from its local x-axis. The entire assembly undergoes $+30°$ rotation about global z-axis. Later, segment 2 rotated by $+45°$ about the local y-axis of the joint. (17) [CO3]

Construct the complete transformation matrix sequence to represent the position of the end-effector tip. Also compute the global (x, y, z) coordinates of the tip.

[Hint: The complete operation can be defined as $P_{ee} = T_{global} \cdot T_{L_1} \cdot T_{joint-operation} \cdot T_{init_1} \cdot T_{L_2} \cdot P_{origin}$ where T's are appropriate transformation matrices.]



6. a) A camera in World Coordinate System (WCS) is defined by eye $(7, 8, 9)$, look-at $(1, 1, 1)$, and up $(0, 1, 0)$. The camera is translated (slid) by $(1, -1, 1)$ in its own coordinate system. Determine the new camera origin in WCS after the slide. (13) [CO3]

b) You are given a point $P(2, 4, 5)$, and a projection plane defined by equation $x - 2y + 2z = 3$. Compute isometric projection of the point on the plane. Also, show the isometric projection matrix. (10) [CO1]

c) Derive the general perspective transformation matrix. Verify that it maps (r, t, n) to $(1, 1, -1)$. Symbols have their usual meaning. (12) [CO2]

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7. a) State how Ward illumination model estimates anisotropic specular reflection. (07) [CO1]

b) A point $P(1,2,3)$ lies on a surface with normal $\boldsymbol{N} = (0,0,1)$, viewer is at $(1,2,8)$. A monochromatic spot light is positioned at $(4,5,6)$ with light focus vector $\frac{1}{\sqrt{3}}(-1,-1,-1)$ and cutoff angle 30°. Determine whether point P lies inside the spot-light cone. If so, compute the angular attenuation factor with fall off exponent = 2. [Note: Be careful about light vector and focus vector's direction.] (08) [CO2]

c) A triangle in 3D space has vertices and normals: $A(1,1,1), N_A(0,-4,3); B(5,1,1), N_B(0,-3,3); C(3,4,5), N_C(1,-4,2)$. A point $P(3,2,2.33)$ lies inside the triangle. Light and view parameters are: L_{pos} $(6,6,8)$, Eye_{pos} $(6,-2,5)$, $I_L = 1$. You are supposed to render the triangle using Phong shading method, and compute intensity by applying Cook-Torrance illumination model. Compute lerped normal at P using Barycentric interpolation. Then, compute specular intensity at P if surface roughness $m = 0.4$, Fresnel base reflectivity $F_0 = 0.1$. Following equations are given for your convenience, where symbols have their usual meaning. Be sure to clamp negative dot products. (20) [CO3]

$$I_{spec} = I_L \cdot f_r \cdot (N \cdot L), \quad f_r = \frac{DFG}{4(N\cdot L)(N\cdot V)},$$

$$\text{Beckmann Distribution } D = \frac{1}{\pi m^2 (N\cdot H)^4} e,$$

$$\text{Schlick Fresnel } F = F_0 + (1 - F_0)(1 - V \cdot H)^5,$$

$$\text{Cook-Torrance Geometry } G = \min\left(1, \frac{2(N\cdot H)(N\cdot V)}{V\cdot H}, \frac{2(N\cdot H)(N\cdot L)}{V\cdot H}\right).$$

8. a) Neatly write shooting algorithm to compute radiosity of a patch. Why does it converge faster than gathering method? (10) [CO1]

b) A tensor product Bézier surface patch $S(u,v)$ is defined as cubic in u direction and quadratic in v direction. (13) [CO3]

i) Write fully expanded matrix equation to compute $S(u,v)$. Your answer must explicitly show the contents and dimensions of vectors U, V, the blending matrices B_i, B_j and geometry matrix P_{ij}.

ii) Using the components from your matrix expression, write the specific 2D basis function $B_{2,1}(u,v)$ that is used for control point $P_{2,1}$. [1D Bernstein polynomial of degree n at index k is $B_{k,n}(t) = \binom{n}{k} t^k (1-t)^{n-k}$]

c) Consider a quadratic B-spline curve (order $k = 3$) is defined by 4 control points $P_0, \cdots, P_3$ and a non-uniform knot vector $T = \{0, 1, 3, 4, 5, 6, 7\}$. Using the Cox-de Boor recursive formula given below, derive the exact algebraic expression for the basis function $B_{1,3}(u)$. (12) [CO2]

$$B_{k,1}(u) = \begin{cases} 1, & \text{if } u_k \le u \le u_{k+1} \\ 0, & \text{Otherwise} \end{cases}$$

$$B_{k,d}(u) = \frac{u - u_k}{u_{k+d-1} - u_k} B_{k,d-1}(u) + \frac{u_{k+d} - u}{u_{k+d} - u_{k+1}} B_{k+1,d-1}(u), \quad d > 1$$

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